13. Cost-effective pelvic floor muscle stimulation for older women

Proposed title: Adaptive Pelvic Floor Rehabilitation and Biofeedback System for Older Women

Clinical need

Discipline: Biomedical engineering + embedded systems + rehabilitation

Problem statement: Older women experiencing urinary incontinence may face barriers to accessing pelvic floor rehabilitation, including cost, discomfort, difficulty performing exercises correctly and limited access to trained professionals. Develop an affordable rehabilitation system that supports individualized pelvic floor muscle training and provides measurable feedback on exercise performance.

Proposed innovation

- A wearable or external biofeedback interface that guides pelvic floor muscle exercises.
- A sensor system to measure appropriate exercise-related signals, such as validated muscle activity where feasible.
- An adaptive training interface that adjusts exercise prompts based on measured performance.
- A session log that helps a qualified clinician monitor progress.

Potential invention hypothesis: A specific closed-loop biofeedback method that adapts exercise guidance to validated muscle-activity measurements while detecting poor signal quality and compensatory muscle activation.

Prototype: Microcontroller, surface EMG acquisition module, reusable external sensors, mobile application and a training dashboard.

Validation: Begin with simulated signals and bench testing. Any clinical evaluation needs appropriate ethics approval and qualified clinical supervision.

Important: electrical stimulation is not suitable for every patient. Do not use a DIY electrical stimulator on participants. Patient-connected stimulation hardware requires electrical safety engineering, clinical oversight and appropriate medical-device regulatory assessment. A biofeedback-only prototype is the safer starting point.

REFERENCE

	Patent	Relevance to your concept	Link
1	**US11426115B2 – System and Method for Pelvic Floor Feedback and Neuromodulation**	Very close prior art: EMG biofeedback for the pelvic floor with sensor electrodes and neuromodulation; addresses electrode placement and equipment limits.	Google Patents – US11426115B2
2	**US11172866B2 – Pelvic Muscle Rehabilitation for Treating Urinary Incontinence**	Garment-integrated EMG sensor measuring pelvic floor contraction plus a moisture sensor, with stored outputs.	Google Patents – US11172866B2
3	**US10602979B2 – Method with Biofeedback for Training the Muscles of the Pelvic Floor**	Biofeedback training method based on EMG signals.	Google Patents – US10602979B2
4	**US6264582B1 – Method for Training of Muscles**	Intravaginal EMG probe, initial assessment of muscle state, auditory feedback and a positive-feedback training programme.	Google Patents – US6264582B1
5	**US6498953B2 – Biofeedback Device for an Incontinent Person**	Senses sphincter muscle electrical activity and signals on a predetermined activity to help the user contract the muscle.	Google Patents – US6498953B2

	Research	What it supports	Link
1	PelviSense, an Innovative, Non-invasive, Biofeedback Device for the Treatment of Stress Urinary Incontinence in Women (NCT06126757)	Registered trial of a reusable wearable EMG sensor with a mobile app showing real-time pelvic floor EMG, comparable to your prototype concept.	ClinicalTrials.gov – NCT06126757
2	Effectiveness and Cost-effectiveness of PelviSense, a Novel Biofeedback Device for Stress Urinary Incontinence in Women (NCT05627726)	Registered trial that includes cost-effectiveness and 7-day progress display in the app.	ClinicalTrials.gov – NCT05627726